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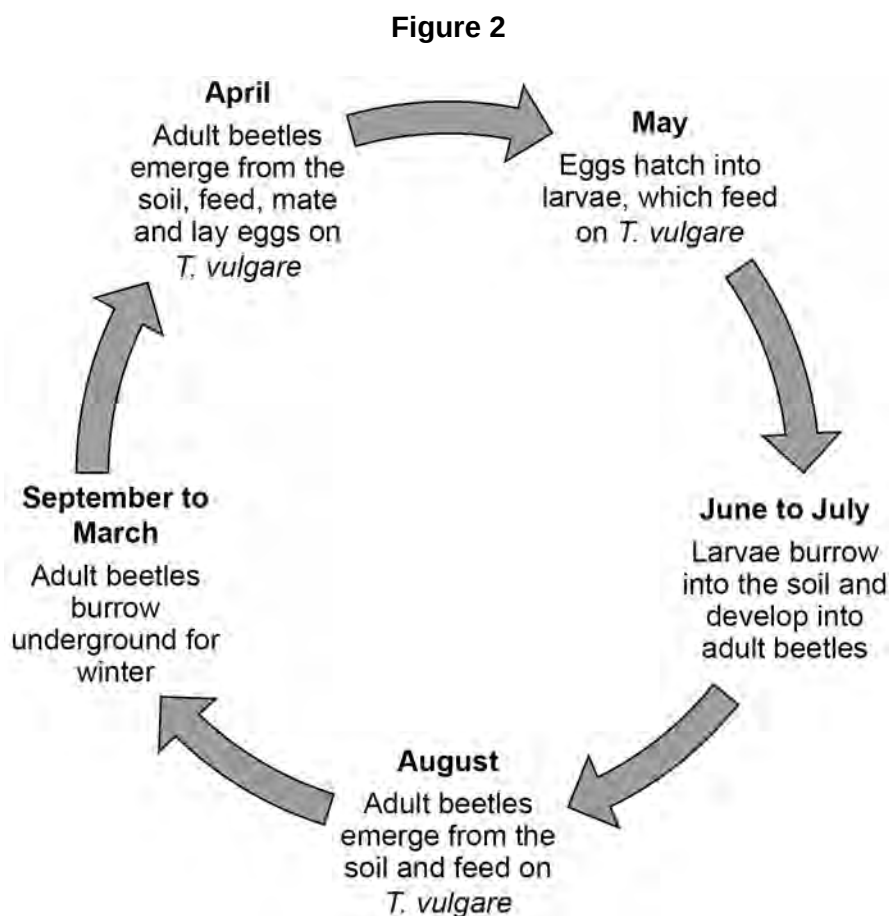
Tansy, *Tanacetum vulgare*, is a plant that grows in dense patches. The tansy beetle, *Chrysolina graminis*, feeds on, and completes its life cycle on *T. vulgare*.

C. graminis is 8–10 mm long and green in colour.

Two other green beetles can also be found in the same habitat.

- The dock beetle, *Gastrophysa viridula*, which is 4–7 mm long.
- The mint leaf beetle, *Chrysolina herbacea*, which is 6–10 mm long.

Figure 2 shows the life cycle of *C. graminis*.



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1

How many different genera of beetles are there in the information provided?

[1 mark]

Question 3 continues on the next page

Turn over ►

Ecologists investigated the population size of *C. graminis* using the mark-release-recapture method.

Table 2 shows the ecologists' results.

Table 2

Number of <i>C. graminis</i> captured and marked in first sample	Number of unmarked <i>C. graminis</i> captured in the second sample	Number of marked <i>C. graminis</i> captured in the second sample
405	248	124

0 3 . 2

The ecologists captured and marked the first sample of beetles during the second week of August. They identified the species of beetle before marking them.

Use the information provided to suggest:

[3 marks]

two reasons why the first sample of beetles was captured during the second week of August

1 _____

2 _____

one reason why the beetles had to be identified before being marked.

1 _____

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Use the data provided in **Table 2** to estimate the population size of *C. graminis*.

Show your working.

[2 marks]

Answer = _____ beetles

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The ecologists repeated the investigation over a five-year period to monitor the population size of *C. graminis*.

State **two** precautions they would have taken each year to obtain comparable data.

[2 marks]

1 _____

2 _____

Turn over for the next question

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